MANIPAL INSTITUTE OF TECHNOLOGY BENGALURU
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
STUDENT MINI-PROJECT WORK DIARY — ACADEMIC YEAR 2026–2027
CLARA: Cloud-Linked Evaluation and Risk Assessment using Predictive Analytics and Business Intelligence

Student Names	1. Johann Mathew Sajan (235890104) 2. Nandan Murali (235890114) 3. Abhijay Gupta (235890206) 4. Arshia Gali (235890070)	**Program / Branch**	B.Tech Computer Science & Engineering (Minor Specialization in Data Analytics)
Section / Roll	AI - A	**Course Code**	CSE 4191 – Minor Specialization Project
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Internal Guide	**Dr. Geetabai S Hukkeri** Assistant Professor, Department of Computer Science and Engineering Dept. of CSE, MIT Bengaluru	**Host Institution**	MIT Bengaluru, BSF Campus, Yelahanka Govindapura, Bengaluru – 560064

Biweekly Period 1 (Weeks 1 – 2) — First Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Literature survey on Cloud Security Posture Management (CSPM) and FinOps standards.	Comprehensive review of CIS AWS Benchmarks and IEEE research.
2	Team charter formulation and project topic finalization with faculty guide.	Scope confirmed for CSE 4191 Minor Specialization in Data Analytics.
3	Initial system requirements analysis and technology stack feasibility study.	Selected DuckDB, scikit-learn, Streamlit, and Google GenAI SDK.

Summary of Work Done	Completed initial project kickoff, established collaborative repository, and finalized core architectural scope with project guide Dr. Geetabai S Hukkeri.
Challenges Faced	Reconciling cloud security misconfigurations with FinOps cost waste in a unified analytical framework.
References Consulted	Sharma et al. (IEEE TCC 2022), Flexera 2024 State of the Cloud Report.
Remarks / Milestones	Project objectives approved; team assigned specialized responsibilities across data, analytics, and UI.
Suggestions by Guide	Ensure synthetic dataset models realistic enterprise multi-cloud topologies rather than single-cloud toy environments.

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Biweekly Period 2 (Weeks 3 – 4) — Second Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Designed normalized 7-table analytical warehouse schema in DuckDB.	Tables: instances, metrics, security_configs, security_events, cost_records, vulnerability_scans, recommendations.
2	Authored synthetic data generation protocol for 300 instances across AWS, Azure, and GCP.	Synthesized 90 days of daily telemetry (27,000 metric rows) using multi-frequency sinusoidal functions.
3	Implemented initial ETL pipeline in data/load_to_duckdb.py with schema validation.	Vectorized CSV ingestion with strict primary/foreign key constraint checking.

Summary of Work Done	Successfully built and validated the data warehouse tier, proving sub-second query performance over 27,000 records.
Challenges Faced	Modeling realistic diurnal and seasonal variance in cloud telemetry without generating unphysical negative values.
References Consulted	Raasveldt & Mühleisen (ACM SIGMOD 2019), CIS AWS Foundation Benchmark v2.0.
Remarks / Milestones	Data warehouse schema design satisfies 3NF normalization while enabling ultra-fast analytical reads.
Suggestions by Guide	Incorporate deliberate security misconfigurations and time-series anomalies into a formal manifest.

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Biweekly Period 3 (Weeks 5 – 6) — Third Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Engineered 15 intentional anomalies across 5 distinct categories into anomaly_manifest.json.	Categories: brute_force, cpu_spike, memory_leak, misconfiguration, network_exfiltration (3 instances each).
2	Quarantined ground-truth manifest from analytical pipeline to eliminate evaluation data leakage.	Enforced strict zero-leakage contract in automated unit tests.
3	Conducted Phase 1 milestone review presentation with department evaluation panel.	Presented architecture, literature survey, and initial warehouse design.

Summary of Work Done	Finalized Phase 1 milestone deliverables, successfully defended project architecture, and established ground-truth evaluation corpus.
Challenges Faced	Ensuring anomaly injection parameters accurately reflect real-world attack signatures and resource leaks.
References Consulted	Patel & Kumar (ACM SoCC 2023), MIT Bengaluru Phase 1 Review Guidelines.
Remarks / Milestones	Phase 1 presentation accepted with appreciation for clear architectural separation of concerns.
Suggestions by Guide	Focus next phase on feature engineering for time-series anomaly detection and explainable posture scoring.

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Biweekly Period 4 (Weeks 7 – 8) — Fourth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Developed statistical time-series feature engineering pipeline in analytics/anomaly_detector.py.	Engineered robust z-score (MAD-based), ratio_7d_30d, rolling regression slope, and peak-to-median ratio.
2	Trained unsupervised Isolation Forest model (scikit-learn) with deterministic seeding.	Configured 100 estimators, max_samples='auto', and active production calibration shift = 0.010.
3	Implemented explainable deterministic security posture scoring model.	Scored encryption status, IAM policy wildcards, public port exposures (SSH 22, RDP 3389), and CVE severity.

Summary of Work Done	Constructed core predictive analytics models for telemetry anomaly detection and security posture quantification.
Challenges Faced	Standard z-scores were heavily skewed by extreme outliers, necessitating robust Median Absolute Deviation scaling.
References Consulted	Liu et al. (IEEE ICDM 2008), Zhang et al. (IEEE BigData 2021).
Remarks / Milestones	Robust statistical scaling resolved outlier distortion and stabilized anomaly scoring.
Suggestions by Guide	Perform honest confusion matrix evaluation against the quarantined manifest.

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Biweekly Period 5 (Weeks 9 – 10) — Fifth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Executed empirical evaluation of Isolation Forest on 300-instance benchmark corpus.	Achieved TP=14, FP=3, FN=1, TN=282 (Precision = 82.35%, Recall = 93.33%, F1 = 0.8750).
2	Conducted honest per-category recall audit across all five anomaly types.	brute_force: 100%, cpu_spike: 100%, memory_leak: 66.7%, misconfiguration: 100%, network_exfiltration: 100%.
3	Verified model generalization across independent random seeds (Seeds 43, 44, and 100).	Confirmed stable precision >= 80% and recall >= 86.6% across unseen datasets.

Summary of Work Done	Completed Phase 2 multi-modal anomaly detection validation, exceeding all mandatory performance criteria.
Challenges Faced	Detecting gradual memory leaks with low daily slope drift without increasing false positive rates on nominal workloads.
References Consulted	Liu, Zhang, & Chen (JNCA 2023), Pedregosa et al. (JMLR 2011).
Remarks / Milestones	Evaluation methodology is academically rigorous; quarantined manifest guarantees zero data snooping.
Suggestions by Guide	Integrate FinOps cost waste quantification and develop the Generative AI recommendation engine.

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Biweekly Period 6 (Weeks 11 – 12) — Sixth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Formulated FinOps cost waste scoring model and rightsizing savings heuristics.	Identified 64 underutilized instances and calculated exactly $220.87 in monthly savings across top 10 recommendations.
2	Designed composite Overall Risk Score formula with strict mathematical parity.	Formula: 0.50 * SecurityScore + 0.30 * AnomalyScore + 0.20 * CostWasteScore.
3	Standardized risk tier classification system across Python, SQL, and UI.	Critical: >= 80, High: [60, 80), Medium: [40, 60), Low: [0, 40), Unknown: invalid/null/NaN scores in neutral gray.

Summary of Work Done	Unified security, anomaly, and cost models into an authoritative composite fleet risk scoring framework.
Challenges Faced	Ensuring mathematical parity between backend scoring calculations and dashboard formula representations.
References Consulted	Kim et al. (IEEE Cloud 2023), Storrs (FinOps Best Practices, O'Reilly 2020).
Remarks / Milestones	Automated regression tests confirm exact weight parity across all application tiers.
Suggestions by Guide	Begin Google Gemini API integration with strict response schema contracts.

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Biweekly Period 7 (Weeks 13 – 14) — Seventh Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Architected production-ready Gemini recommendation engine in genai/gemini_client.py.	Integrated google-genai SDK with configured model string gemini-3.8-flash.
2	Enforced strict Pydantic v2 schema contracts (RecommendationBatch and RecommendationItem).	Prevented LLM hallucination of schema fields, priorities, and instance identifiers.
3	Constructed autonomous deterministic offline fallback engine in genai/recommendation_engine.py.	Implemented rule-based expert engine executing in <5ms during API outages or missing keys.

Summary of Work Done	Engineered the dual-path recommendation system, achieving full operational decoupling between cloud API and core scoring.
Challenges Faced	Handling non-deterministic JSON formatting and markdown code-block wrapping from foundation model outputs.
References Consulted	Zhao & Park (ACM SIGMOD 2023), Google DeepMind Gemini Technical Report 2023.
Remarks / Milestones	Dual-path architecture guarantees high reliability: system never crashes on missing API keys or network partitions.
Suggestions by Guide	Implement authoritative, anti-spoofing provenance controls to prevent model identity spoofing.

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Biweekly Period 8 (Weeks 15 – 16) — Eighth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Implemented application-controlled provenance attribution in genai/recommendation_engine.py.	Stripped source/model fields from LLM schema; application state authoritatively assigns provenance.
2	Executed live Gemini API smoke test using local environment key.	Returned 10 verified recommendations with generation_source='gemini', model_name='gemini-3.8-flash', fallback_reason=NULL.
3	Added comprehensive fault-injection test suite in tests/test_recommendations.py.	Verified automatic fallback on missing keys, auth errors (401), timeouts, and quota exhaustion (429).

Summary of Work Done	Successfully validated live Gemini integration and proved cryptographic provenance integrity in DuckDB.
Challenges Faced	Preventing Gemini responses from falsely claiming offline fallback or alternative model names.
References Consulted	Wang et al. (ICSE 2024), NIST SP 800-145 Cloud Computing Guidelines.
Remarks / Milestones	Live smoke test verified end-to-end functionality; all 10 recommendations link to valid instance records.
Suggestions by Guide	Proceed to Streamlit dashboard development and usability optimization.

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Biweekly Period 9 (Weeks 17 – 18) — Ninth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Constructed interactive multi-page Streamlit dashboard in dashboard/app.py and dashboard/pages/.	Built 5 dedicated pages: Welcome, Executive Overview, Security Analysis, Cost Optimization, AI Recommendations.
2	Engineered responsive Plotly data visualizations.	Implemented fleet risk donut chart, 90-day cost trend projection, posture score histogram, and utilization scatter plot.
3	Implemented centralized data access layer with connection pooling in dashboard/utils/data_access.py.	Optimized SQL analytical queries to execute in <15 milliseconds with graceful empty-state handling.

Summary of Work Done	Delivered executive-grade business intelligence dashboard delivering real-time cloud governance visibility.
Challenges Faced	Eliminating chart rendering flickers and persisting filter selections across multi-page navigation.
References Consulted	Streamlit Documentation v1.30, Plotly Open Source Graphing Libraries.
Remarks / Milestones	Dashboard UI is responsive, aesthetically polished, and adheres to dark/light theme best practices.
Suggestions by Guide	Conduct rigorous end-to-end testing and verify dashboard calculations against backend models.

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Biweekly Period 10 (Weeks 19 – 20) — Tenth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Developed comprehensive pytest test suite across 75 test cases in tests/ directory.	Achieved 100% test pass rate across data quality, ML models, GenAI contracts, and dashboard UI logic.
2	Enforced strict test isolation using session-scoped temporary directory fixtures.	Guaranteed that production warehouse (clara.db) remains bit-for-bit unchanged during test runs.
3	Eliminated unverified operational claims and aligned narrative with empirical evidence.	Replaced CVSS claims with observable indicator scoring; verified exact $220.87 cost savings.

Summary of Work Done	Achieved complete test coverage and established rigorous automated verification across the entire platform.
Challenges Faced	Ensuring hermetic database isolation in multi-threaded pytest execution environments.
References Consulted	pytest Framework Documentation v8.2, IEEE Standards for Software Testing (IEEE 829).
Remarks / Milestones	Test suite runs cleanly in under 70 seconds; zero failing or flaky tests.
Suggestions by Guide	Prepare comprehensive academic technical report and presentation deck according to MAHE guidelines.

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Biweekly Period 11 (Weeks 21 – 22) — Eleventh Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Authored 38-page comprehensive academic thesis report complying with official Phase 2 UG template.	Included formal front matter, 10 detailed chapters, 20 IEEE references, and NBA/IET articulation matrices.
2	Formulated NBA Course Articulation Matrix (Table A1.1, A1.2) and IET CLO-AHEP Mapping (Table A1.3).	Mapped CO1-CO5 to engineering program outcomes (PO1-PO12) and accreditation criteria.
3	Engineered high-DPI system architecture and data pipeline diagrams in docs/final/figures/.	Rendered 7 publication-quality vector/PNG diagrams illustrating core workflows.

Summary of Work Done	Completed formal academic thesis documentation, exceeding the mandatory 35-page threshold with deep technical exposition.
Challenges Faced	Structuring complex multi-modal algorithms, mathematical proofs, and schema contracts into cohesive thesis chapters.
References Consulted	MAHE Phase-2 UG Thesis Guidelines, NBA Accreditation Manual for Undergraduate Programs.
Remarks / Milestones	Academic report is exceptionally thorough; mathematical formulations and empirical tables are fully documented.
Suggestions by Guide	Develop final viva presentation slides, presentation speech script, and final packaging.

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Biweekly Period 12 (Weeks 23 – 24) — Twelfth Fortnight

Sl.	Details of Work Carried Out	Comments / Verification Status
1	Constructed 18-slide executive presentation deck (PPTX and PDF) with embedded presenter notes.	Covered architecture, methodology, empirical evaluation, live Gemini validation, and demo walkthrough.
2	Authored comprehensive 20-minute presentation spoken transcript in docs/final/CLARA_Presentation_Script.docx.	Embedded slide-advance cues, timing allocations, and transition phrasing.
3	Assembled and verified biweekly project diary with student and faculty guide signature blocks.	Synchronized all 12 biweekly work entries across project lifecycle.
4	Executed automated secret scan across repository files, confirming zero leaked credentials.	Excluded virtual environments, local caches, and private environment files from submission manifest.

Summary of Work Done	Finalized all Phase 2 deliverables, presentation assets, project diary, and verification benchmarks for final examination.
Challenges Faced	Synthesizing extensive experimental results into a concise, impactful 20-minute viva presentation format.
References Consulted	MIT Bengaluru CSE 4191 Project Evaluation Criteria, IEEE Presentation Guidelines.
Remarks / Milestones	All Phase 2 deliverables are complete, verified, and ready for departmental evaluation.
Suggestions by Guide	Review viva defense questions and ensure all team members are prepared for technical examination.

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